

C-MCPN: The Clingo McCulloch-Pitts Network Editor

An Answer-Set Programming Framework for Artificial Neural
Networks

David Gonzalez¹, Joshua T. Guerin, Ph.D.²

Mathematical Sciences

University of Northern Colorado

Greeley, CO 80639

¹*****@bears.unco.edu

²*****@unco.edu

Abstract

As Artificial Intelligence and generative AI continue to engage the public in ways that mirror other critical advances (e.g., the PC and smartphone revolutions as well as the broader Internet) public interest has turned more and more towards applications of the underlying technologies and architectures that power them. While machine learning technologies like LLMs have a profound capacity to integrate AI into modern life, the underlying artificial neural network technologies present significant concern for both short and long-term sustainability: data center footprint, waste heat and noise, and infrastructure/resources for energy and clean water.

At the same time, advanced declarative languages have made remarkable headway towards AI solutions are considerably more efficient. In particular, Answer Set Programming languages are designed to solve computationally difficult (e.g., NP-Complete and NP-Hard) problems with limited available resources. In this paper we outline how these languages can be used to model artificial neural network architectures, like those used in LLMs, in the modeling language Clingo. We also introduce the C-MCPN (Clingo McCulloch-Pitts Network) Framework and Editor—a tool largely written in Python and Clingo. C-MCPN was developed to support our work in generating more efficient neural networks by providing a graphical interface to work with these models, using Clingo as a back-end for both representation and inference.

1 Introduction

Artificial Intelligence (AI) systems and their applications have been extremely consequential on the modern world over the decades. This has been especially evident over the last decade with skyrocketing awareness and use of generative AI (GEN-AI) techniques. In particular, these include techniques from machine learning (ML) like the artificial neural networks (ANNs) that power contemporary large language models (LLMs).

These AI systems have shown great promise, but the underlying technologies come with significant drawbacks. Concerns include the growing footprint of data centers, which has garnered public interest [13]; the worsening use of finite resources such as fresh water and electricity [10]; and the potential for substantial environmental and health impacts [8].

Conversely, Answer Set Programming (ASP) languages live at a different end of the research spectrum in terms of their efficiency. They are designed to solve NP-Complete and NP-Hard problems in a fast and efficient manner on limited hardware. Our current work involves exploring the use of ASP languages to accelerate the processes of learning and inference. With this, we hope to reduce the resources needed for these technologies.

In this paper we introduce the **C-MCPN Editor**: The *Clingo McCulloch-Pitts Network Editor*. The C-MCPN Editor is a tool we developed to assist in the exploration of ASP-based networks. It allows a user to model the network architecture, demonstrating that ANNs, specifically McCulloch-Pitts Networks (MPNs), can be used to model and conduct inference over McCulloch-Pitts neurons [11].

2 Background

Our work is related to the merging of ANN technology with more efficient Answer Set Programming languages. In this section we will provide a brief background on both subjects, motivating our work on C-MCPN.

2.1 Artificial Neural Networks (ANNs)

ANNs were originally inspired by the structure and function of biological neurons. Their basic building blocks, the artificial neurons, can be traced back to the work of McCulloch and Pitts in the 1940s [11]. These early models mimicked the behavior of biological neurons by using a simple thresholding function that determines whether a neuron will fire or not. The name for the C-MCPN was inspired by this early work, though its implementation is not a pure McCulloch-Pitts model. Instead, it uses a more generalized version

of neuron with two functions: an excitation function $g(\vec{x})$ and an activation function $f(g(\vec{x}))$, which can be seen in Figure 1.

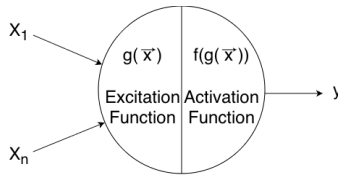


Figure 1: A diagram of a 2-input artificial neuron

These neurons can be connected together to form a network, allowing for the processing of complex data and the learning of patterns through training. The structure of these networks can vary widely, from simple knowledge graphs to feedforward networks to more complex recurrent architectures [18]. For our work, the feedforward neural network was chosen as the basis because of its simplicity and wide use in the field. These consist of layers of interconnected neurons that process data in a single forward direction [7]. An example of a small network of this type can be seen in Figure 2. These models can learn through a training process where weights are adjusted based on error, but notably their underlying structure is very similar to the original McCulloch-Pitts model.

ANNs have been widely adapted in various applications, from image recognition to natural language processing like the transformer-based models such as BERT and GPT [16, 5, 2]. However, it is no secret that they have come with significant drawbacks. Among these is the issue of resource use, discussed further in Section 2.1.1. This is far from the only concern, but it is the one that most motivates our work.

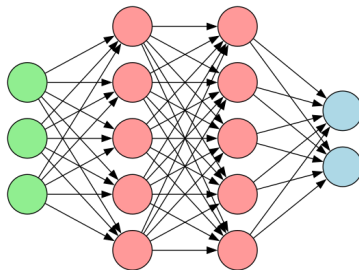


Figure 2: A diagram of a simple feedforward neural network

2.1.1 Environmental Impacts

The introduction of big data, chiefly by scraping the broader Internet, changed the landscape for AI. It allowed for statistical-based models to flex their strength, which require massive amounts of data for training purposes [9, 1]. This incredible need for information highlights an important drawback in the form of *data efficiency* [4]. The lack of efficiency then translates to a significant amount of necessary resources.

As an example, consider the cost to train a single model, ChatGPT-3. According to Vries [17], GPT-3 reported around 1.3 million kWh *just* to be trained. With some quick, back-of-the-envelope calculations, we can put this into perspective. In 2022 the average American household used approximately 10,791 kWh [15], and by dividing these out we find that the model’s energy use was proportional to:

$$1,300,000 \text{ kWh} \times \frac{1 \text{ household}}{10,791 \text{ kWh/year}} \approx 120.5 \text{ households.}$$

Similarly, we could produce a rough, *lower bound* of how much water may have been used for training. A thermoelectric power station can use approximately 43.8 L/kWh of water [10]. That would put the *indirect* water costs of training GPT-3 at approximately 56,940,000 L, approximated by:

$$1,300,000 \text{ kWh} \times 43.8 \frac{\text{L}}{\text{kWh}} = 56,940,000 \text{ L}$$

Given that an adult male averages approximately 4L of fresh water [12], that number represents sufficient water to sustain a lower bound of approximately 39,000 adults for a year:

$$56,940,000 \text{ L} \times \frac{1 \text{ adult}}{4 \text{ L/day} \times 365 \text{ days}} \approx 39,000 \text{ adults}$$

That amount is sufficient to provide fresh water to a city of approximately the size of Brighton, CO for a year. Though these calculations include *only* the indirect costs and are remarkably overly-simplified, we hope that it puts into perspective how the technology is already unsustainable. This highlights the dire need for more efficient, sustainable AI and ML technologies [17], an open problem yet to be solved.

2.2 Answer Set Programming (ASP)

Answer Set Programming serves as an amalgamation of Prolog-style quantified logic with powerful advances in satisfiability (SAT) solvers that occurred

throughout the 1990s and into the 2000s. In this section we will discuss the basic syntax and semantics of the ASP language, Clingo, which serves as a technological foundation of C-MPCN [6, 3].

2.2.1 Syntax and Semantics

The atomic units in Clingo are *literals*, which describe ideas and *facts* that serve as statements that are held to be true. The basic syntax is the application of a *property* to a literal, `property(atom)`. E.g.,

```
red(apple).
```

is a complete statement (and program), attributing a property of red to an idea named `apple`.

Further knowledge is *inferred* using aggregate statements known as *rules*, taking the form: `consequent :- antecedent`. This mirrors an *implication* from formal logic, as the `:-` symbol is read equivalently to $\leftarrow$. In such a statement the consequent *must* be true whenever the antecedent is true. E.g.,

```
eat(Fruit) :- red(Fruit), ripe(Fruit).
```

describes a preference over what fruit will be eaten. I.e., “If a fruit is red and ripe, then we eat the fruit.”

In this case rather than the literal `apple`, we use a *variable* to allow for inference. The `Fruit` can be *any* contextually-relevant literal in the program, so long as it possesses the required properties.

3 Methodology

Combining the efficiency of ASP languages with the representational power of ANNs is a non-trivial task. Not only does it require a novel representation of the ANN structure, but Clingo has no floating-point support. This requires clever simplifications of the activations functions and gradient descent to use them over integers. Additionally, large ANNs can cause combinatorial explosions in the search space, making grounding the programs difficult. The C-MPCN Framework and Editor were developed to address some of these challenges, providing a structured way to represent and work with logical networks in Clingo.

3.1 Domain Selection

To build and test the framework the illustrative domain of digital logic for hardware development was selected. Its simplicity and familiarity make it an ideal domain for experimentation.

For our example network, a 1-bit adder seen in Figure 5, we used neurons with three types of well established logic gates: AND, OR, and XOR. In Table 1 we show the McCulloch-Pitts definitions of these gates, and their truth tables. Using a combination of these and more gates, one is able to construct complex logical operations¹.

Table 1: McCulloch-Pitts Neuron Definitions and Truth Tables.

AND			OR			XOR		
$g(\vec{x}) = x_1 \cdot x_2, \theta = 1$			$g(\vec{x}) = x_1 + x_2, \theta = 1$			$g(\vec{x}) = x_1 - x_2 , \theta = 1$		
Activation:			$f(\vec{x}) = \begin{cases} 0 & \text{if } g(\vec{x}) < \theta, \\ 1 & \text{if } g(\vec{x}) \geq \theta. \end{cases}$					
AND			OR			XOR		
x_1	x_2	$f(\vec{x})$	x_1	x_2	$f(\vec{x})$	x_1	x_2	$f(\vec{x})$
T	T	T	T	T	T	T	T	F
T	F	F	T	F	T	T	F	T
F	T	F	F	T	T	F	T	T
F	F	F	F	F	F	F	F	F

3.2 Design Approach

The representation of an artificial neuron requires both the *structure* designation, and the function used to *activate* it. When activated the neuron will propagate a signal to its output given the provided input signals. Its value then further propagates through the network along edges and activating other neurons.

An AND gate is used to illustrate how we represent an artificial neuron:

```
% gate(TYPE, OUTPUT, INPUTS)
gate("AND", Out, In1, In2) :- values(In1), values(In2),
                               Out = (In1 * In2).
```

The *structure* of the neuron is indicated by the parameters: in the case of an AND gate this requires two input parameters and a single output. The *activation function* is encoded directly in the rule structure using the arithmetic equivalent provided in Table 1.

¹An elegant and simple proof of Clingo's Turing completeness falls out of the C-MCPN Framework.

The neuron *connectivity* specified in the gate definitions can then be used to forge connections of a network. A connection from the 1-bit adder sample network is built as:

```
% neuron(type, unique_id)
neuron("INPUT", "INPUT_1").
neuron("XOR", "XOR_1").

% edge(source_id, target_id)
edge("INPUT_1", "XOR_1").

% val(neuron_id, value).
val("INPUT_1", 1).
```

Finally, the value inference program is used to store the possible values, activate the neurons, and propagate the values through the network. The program for a 2-input gate can be built as:

```
%% Value domain
values(0;1).

% Edges carry values from input to output
edge(In, Neuron_id, Val) :-
    edge(In, Neuron_id),
    neuron(_, Neuron_id),
    neuron(_, In),
    val(In, Val).

% 2 input gates
val(Neuron_id, Val) :-
    gate(Type, Val, Val1, Val2),
    neuron(Type, Neuron_id),
    edge(In1, Neuron_id, Val1),
    edge(In2, Neuron_id, Val2),
    In1 != In2.

%% Each neuron has one value
{ val(Neuron_id, Val) : values(Val) } 1 :- neuron(_, Neuron_id).

%% Show values
#show val/2.
```

The rule carrying the values along the edges is vital to the functioning of the network. Without it, the network gets "stuck" and is unable to infer what the values should be. This is a key insight we had during development.

4 C-MCPN

To better explore the ANNs implemented within Clingo we developed the C-MCPN software package to automate our workflow. The system itself is designed around a standard (unix-like) terminal use and is extended into a graphical user interface (GUI).

The core functionalities of C-MCPN are built on the ASP language Clingo, as described in the preceding sections. The editor and interface were constructed in Python using the DearPyGui library, allowing us to take advantage of its node-based editing capabilities.

For node reorganization in the editor we used a modified version of the Sugiyama algorithm for layered graph drawing, taking just the layer assignment step and a simplified node positioning step [14].

4.1 C-MCPN Framework

From what the authors found in the literature, the C-MCPN Framework is a novel approach to neural network representation within a logical programming paradigm like ASP. The networks themselves have relatively simple representations that can be leveraged to create complex operations. The framework, which can be seen in Figure 3, uses three main program types, all of which are contained in the C-MCPN/ directory of the project:

- **Gate Definition Programs**
- **Network Structure Programs**
- **Value Inference Program**

The content in the gate definitions program defines the behavior of each gate, but every default gate we included also has a special header used to send meta-information to the C-MCPN Editor.

4.2 C-MCPN Editor

The `app/` directory of the project contains the source code for the C-MCPN Editor, which can be thought of as an interactive development environment (IDE) for logical networks. Its main features include the ability to add/remove neurons, connect/disconnect them, copy/paste them, and run inference on the network. Additionally, the editor provides a live preview of the ASP code generated from the network structure, allowing users to see how their changes affect the underlying logic program.

User created networks are saveable to JSON and ASP format, and loadable from JSON format. This facilitates creation of complex ANNs, and saving the

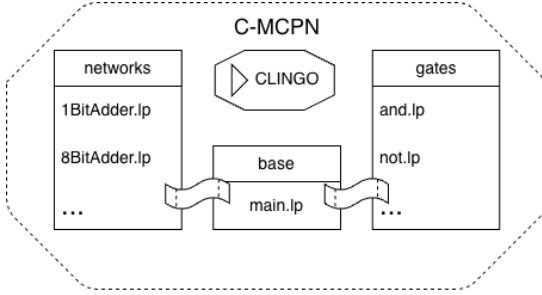


Figure 3: A visualization of the programs involved in the C-MCPN Framework.

network to ASP format allows direct use of the C-MCPN Framework from the terminal instead of the IDE.

A screenshot of the editor can be seen in Figure 4, showcasing the sample 1-bit adder network. A closer look at the sample network can be seen in Figure 5.

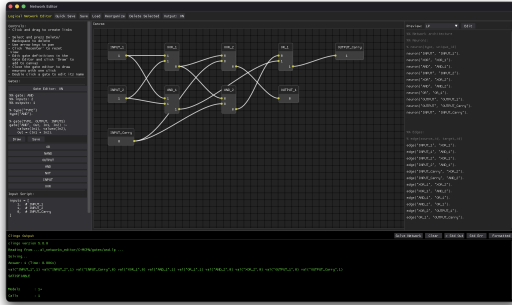


Figure 4: A screenshot of the C-MCPN Editor, showing a sample 1-bit adder network.

4.3 AI-Assisted Development

Claude (Anthropic) was used as a programming assistant throughout development of the editor, helping with DearPyGui module integration, debugging, and some code generation. All design decisions, research direction, and domain-specific logic (ASP, gate definitions, network structure) were conscious decisions by the authors. Each line of code generated by Claude was reviewed, edited, and approved by the authors before being added to the codebase. In

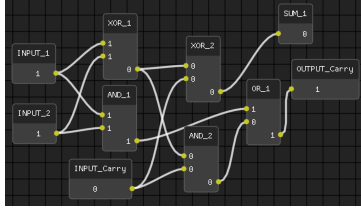


Figure 5: A screenshot of a 1-bit adder network in the C-MCPN Editor.

order to maintain a supervisory-level of control the agent was not given direct access to the codebase or the repository. The files that Claude had input in are contained in the `app/` directory of the project, allowing users the ability to inspect these files if they so desire.

Use of Claude was employed only in the above described sections in order to accelerate the development process under circumstances of limited resources.²

5 Results

The results of this work is the C-MCPN Framework and Editor, which provide a novel approach to representing and working with logical networks within the paradigm of logic programming. The framework and included neuron gates were validated for correctness using an 8-bit adder network, and the editor was used to construct and visualize this network. While being the main motivation for this work, results on its efficiency and scalability are left for future work since that will matter more for the adaptive neural network framework we are currently developing. The C-MCPN Framework and Editor are both open source and available at https://github.com/*****.

5.1 Limitations

This work is a proof of concept for building ANNs in Clingo, and as such there are some limitations to be aware of. The framework has no cycle detection nor has it been tested on recurrent networks. This is likely to cause issues with the value inference program, as it relies on the values being propagated along the edges. The framework has also not been tested on large networks, and it is unknown how scale will affect its performance. Also, as noted in Section 3, Clingo does not have native support for floating point numbers, limiting its potential applications.

²The authorship of this paper is without the assistance of any AI system, including planning, writing, editing, and repository maintenance.

5.2 Future Work

While there are many potential avenues for future work in this area, we are most excited about the possibility of using the C-MCPN Framework as the skeleton for an adaptive neural network which can learn on data. This framework is in its early stages, but we hope to also add it as a feature in the C-MCPN Editor in the future.

6 Conclusion

In this paper, we presented the C-MCPN Framework and Editor, demonstrating that Answer Set Programming can be used to represent and conduct inference over artificial neural network structures, specifically McCulloch-Pitts Networks. Our contribution is twofold: a novel two-layered representation for encoding ANN structure and inference in ASP, and a working, accessible tool that makes this representation practical to construct, visualize, and experiment with. Together, these establish a foundation for future work exploring whether ASP-based learning models can offer meaningful efficiency gains over traditional ANN implementations, an open question we are actively pursuing.

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